

## Mukundkumar Patel

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### EDUCATION:

**Montclair State University** (GPA: 4.0/4.0) September 2024 - May 2026

Master of Science: Data Science

**CVM University** (GPA: 3.58/4.0) June 2020 - March 2024

Bachelor of Engineering: Information Technology

### TECHNICAL SKILLS:

- Programming languages: Python, SQL, R, DAX, Power Query
- Tools: MS Power BI, MS Excel (VLOOKUP, Pivot Tables), GitHub, ArcGIS, Tableau, dbt, Looker Studio
- Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, ArcPy, Folium, Plotly, Google Gemini, Anthropic API
- Database and Cloud: MySQL, Oracle, SQLite, Data Lake, PostgreSQL, MS Access, Google Cloud

### WORK EXPERIENCE:

**Data Analyst Intern** (MTA - Metropolitan Transportation Authority) October 2025 - May 2026

- **Automated data processing** and data validation of vendor-provided data for 8,500+ events **using Python and Office Scripts**, identifying and correcting inaccuracies to improve overall data quality.
- Managed and improved master data for 1,400+ bus routes using **SQL** and Excel; performed **ArcGIS-based spatial analysis** to identify route overlaps and improve operational efficiency.
- Designed a Python computer vision pipeline to detect bus lane and bus stop violations from 900+ traffic cameras, achieving 92% detection accuracy; integrated results with **Power BI** dashboards to visualize violation trends for accurate ticket issuance.
- Built a Python **ETL workflow** to process driver trip logs, calculating pull-in/pull-out counts per depot at 30-minute intervals to optimize fleet planning, maintenance schedules, and backup coordination.

**Graduate Research Assistant** (Montclair State University) August 2025 - May 2026

- Standardized and governed institution-wide student **master data** (8,000+ records) using **SQL** and spreadsheets for federal compliance, institutional research, and regulatory reporting.
- Created **Tableau** dashboards to analyze 5+ years of student data across enrollment, financial aid, and academic outcomes; presented KPI trends and performance reporting insights to stakeholders for leadership review.

**Data Analyst** (Montclair State University) April 2025 - August 2025

- Streamlined the data cleaning process, **reducing processing time by 30%**, and structured raw data into categorized community partners, focus areas, and population segments for targeted impact analysis.
- Developed an interactive **Python map** and Power BI dashboards to analyze 150+ community investment activities, visualize top-performing initiatives, track month-wise trends, and identify key regions for targeted investment.

### PROJECT DETAILS:

**E-Commerce Customer Analytics Pipeline** ([GitHub](#)) ([Dashboard](#)) May 2026

- Built a **5-model dbt pipeline** on **BigQuery** covering GA4 event schema modeling, behavioral segmentation across 269K users, and cross-source identity resolution, replicating core Segment **CDP** functionality.
- Identified 9,630 cart abandonment targets via **GA4 UNNEST** queries on nested event\_params; simulated Klaviyo-style recovery campaign using date-split A/B methodology, projecting **23% recovery rate** vs 8% baseline and \$25K incremental revenue; visualized results in **Looker Studio** dashboard.

**Revenue Intelligence Agent** ([GitHub](#)) ([Dashboard](#)) April 2026

- Built autonomous multi-agent AI pipeline using **Groq LLM** and Python, monitoring competitor car rental pricing across 8 competitors and 10 US airports by integrating real-time flight, event, and weather demand signals, reducing manual revenue analysis from 3 hours to 15 minutes via **GitHub Actions** automation.
- Developed end-to-end pricing intelligence system with **REST API** integration, **SQLite database**, and interactive **5-tab Streamlit** dashboard, automatically flagging **competitor price anomalies  $\geq 20\%$**  and generating urgency-ranked pricing recommendations across economy, compact, and SUV categories.

### EXTRACURRICULAR ACTIVITIES:

**Research Volunteer** (Montclair State University) November 2024

- Processed NJ weather data (1915-2024) and contributed to research showing ARIMA, SARIMA models outperform deep learning methods in time and space efficiency for time-series forecasting.

### CERTIFICATIONS:

- Data Science Fundamentals for Data Analysts by Databricks ([Certificate](#))
- Microsoft 365 Fundamentals ([Certificate](#))
- Machine Learning with Python by IBM ([Certificate](#))
- Introduction to Applied Business Analytics by University of Illinois ([Certificate](#))